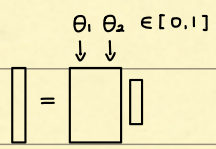
$y_M = A_{M\times N} x_N + n_M$ 无格点，θ 之间可无限近



$a(\theta) = [\, e^{j2\pi d_0\theta} \cdots e^{j2\pi d_{M-1}\theta} \,]$

超分辨问题：准确恢复 Separation Δ 能多小

准确恢复也需 Valid criteria，CD kernel 给了 $L \to \infty$ 时的 criteria

Lemma 1. 准确恢复的充分条件

$$K_M^{-1}(\zeta,\zeta) \longrightarrow \mu(\{\zeta\}), \quad \forall |\zeta| = 1 \qquad (9)$$

满足 (9) 能说明 $M \to \infty$ 时准确恢复

若 Δ 太小 (9) 不成立　本文看用这种分析 Δ 能多小 (Q)

测度 $\mu(\theta) = \sum p_n \mathbb{1}(\theta - \theta_n) + \sigma^2\theta$

$$d\mu(\theta) = \sum p_n \delta(\theta - \theta_n) + \sigma^2 d\theta$$

矩 $m_k = E[e^{j2\pi\theta k}] = \int_{\theta\in[0,1]} e^{j2\pi\theta k} d\mu(\theta) = \begin{cases} \sum p_n + \sigma^2, & k = 0 \\ \sum p_n e^{j2\pi k\theta_n}, & \text{else} \end{cases}$

$m_k = \mathrm{cov}(y_i, y_{i+k})$

矩的 Toeplitz 矩阵 T　Covariance R　(ULA)

CD kernel　$a^H R^{-1} a$

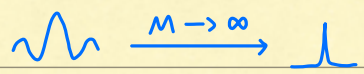
$$K = \psi(z)^H T^{-1} \psi(s) \qquad \psi(z) = \begin{bmatrix} 1 \\ z \\ \vdots \\ z^{M-1} \end{bmatrix}$$

MVDR　$\dfrac{1}{a^H R^{-1} a}$

ψ　T

Szegö Extremum：准确恢复的充分条件

$$\lim_{M\to\infty} K^{-1}(\zeta,\zeta) = \mu(\{\zeta\}), \quad \forall |\zeta| = 1$$



Dirichlet

$$D_M(\pi x) = \frac{\sin(M\pi x)}{\sin(\pi x)}$$

Fejer Kernel

$$F_M(x) = \frac{1}{M} D_M^2(\pi x) = \frac{\sin^2(M\pi x)}{M\sin^2(\pi x)}$$

粗估计

$$F_M(x) \le \frac{1}{M} \min\left\{ M^2 \cdot \frac{1}{4x^2} \right\}$$

用 θ_1、θ_2 表示 K

$$K(e^{j2\pi\theta_1}, e^{j2\pi\theta_1}) = a^H(\theta)T^{-1}a(\theta) = \frac{M}{\sigma^2}(1-\gamma_1\gamma_2)$$

$$\gamma_1 = 1 + \frac{1}{M}\frac{\sigma^2}{P_2} - \frac{1}{M}F(\Delta) + \frac{1}{M^2}\frac{\sigma^2}{P_1}F(\Delta)$$

$$\gamma_2 = \left[1 + \left(\frac{\sigma^2}{P_1} + \frac{\sigma^2}{P_2}\right)\frac{1}{M} + \frac{\sigma^4}{P_1P_2}\frac{1}{M^2} - \frac{1}{M}F(\Delta)\right]^{-1}$$